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WILSON, SONSINI, GOODRICH & ROSATI			EXAMINER	
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Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

patentdocket@wsgr.com

DETAILED ACTION***Comments***

The present application is being examined under the pre-AIA first to invent provisions. In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

Claims 1-20 are pending and examined in the instant Office action.

Terminal Disclaimer

The terminal disclaimer filed on 23 May 2019 disclaiming the terminal portion of any patent granted on this application which would extend beyond the expiration date of U.S. Patent 10,068,051 has been reviewed and is accepted. The terminal disclaimer has been recorded.

Withdrawn Rejections

The double patenting rejections are withdrawn in view of the terminal disclaimer filed on 23 May 2019.

Priority

The instant application is a continuation of 14/451,876, filed 8/5/2014. Application '876 is a continuation of 13/756,760, filed on 2/1/2013. Application '760 claims benefit to provisional applications 61/703,093, filed on 9/19/2012 and 61/594,480, filed 2/3/2012. The provisional applications do not disclose the independent claim limitation of partitioning the mixture into a plurality of droplets. Consequently, the benefit date for claims 1-20 is 2/1/2013.

Claim Rejections - 35 USC § 103

The following is a quotation of pre-AIA 35 U.S.C. 103(a) which forms the basis for all obviousness rejections set forth in this Office action:

(a) A patent may not be obtained though the invention is not identically disclosed or described as set forth in section 102, if the differences between the subject matter sought to be patented and the prior art are such that the subject matter as a whole would have been obvious at the time the invention was made to a person having ordinary skill in the art to which said subject matter pertains. Patentability shall not be negated by the manner in which the invention was made.

This application currently names joint inventors. In considering patentability of the claims under pre-AIA 35 U.S.C. 103(a), the examiner presumes that the subject matter of the various claims was commonly owned at the time any inventions covered therein were made absent any evidence to the contrary. Applicant is advised of the obligation under 37 CFR 1.56 to point out the inventor and invention dates of each claim that was not commonly owned at the time a later invention was made in order for the examiner to consider the applicability of pre-AIA 35 U.S.C. 103(c) and potential pre-AIA 35 U.S.C. 102(e), (f) or (g) prior art under pre-AIA 35 U.S.C. 103(a).

The following rejection is reiterated:

35 U.S.C. 103 Rejection #1:

Claims 1-12 and 20 is/are rejected under pre-AIA 35 U.S.C. 103(a) as being unpatentable over Han et al. [Nature Biotechnology, volume 19, 2001, pages 631-635; on IDS] in view of Lin et al. [WO 2012/058638 A2] in view of Vogelstein et al. [PNAS, volume 96, 1999, pages 9236-9241].

Claim 1 is drawn to a method of detecting any unique combination of presence or absence of at least five polynucleotides in a plurality of droplets. The method comprises providing a sample potentially comprising at least one of the at least five polynucleotide analytes. The method comprises forming a mixture of the sample and at least five hybridization probes, each comprising a fluorophore selected from at most four fluorophores. The method comprises partitioning the mixture into the plurality of droplets. The method comprises exciting the fluorophores to generate signals of one or more of the at least five polynucleotide analytes is present in said plurality of droplets. The signals comprise at least one signal generated by excitement of the one or more fluorophores. The method comprises measuring the signals to generate a cumulative measurement. The cumulative measurement corresponds to the presence of a unique combination of presence or absence of the at least five polynucleotide analytes in the sample. The method does not require any step of immobilization of the at least five polynucleotide analytes or mass spectrometry.

Claim 2 is further limiting wherein each of the at least five hybridization probes is complementary to one of the at least five polynucleotide analytes.

Claims 3-6 and 8-9 provides further constraints on the numbers of analytes, probes, and fluorophores.

Claim 7 is further limiting and requires PCR.

The document of Han et al. studies quantum dot tagged microbeads for multiplexed optical coding of biomolecules [title]. Figure 5 on page 634 of Han et al. illustrates four different polynucleotide analytes labeled with four different combinations of three different fluorophores. The bottom three polynucleotide analytes are hybridized to complementary polynucleotide probes. When the sample undergo spectroscopy, the optical code on the right panel of the figure is the cumulative measurement of the combination of fluorophores for each of the four polynucleotide analytes. The technique of Han et al. does not involve immobilization or mass spectrometry.

Han et al. does not teach a mixture of at least five hybridization analytes. Han et al. does not teach partitioning the mixture into droplets or conducting PCR.

The document of Lin et al. teaches nucleic acid nanostructure barcode probes [title]. The abstract of Lin et al. teaches that uniquely fluorescently labeled polynucleotide analytes are used as barcode hybridization probes. Figure 2 of Lin et al. illustrates histograms for a combinatorial number of probes having one of 27 different barcodes.

Han et al. and Lin et al. do not teach partitioning the mixture into droplets or conducting PCR.

The document of Vogelstein et al. studies digital PCR [title]. The methods section of Vogelstein et al. teaches diluting and partitioning a genome into many small droplets on many 96 well plates such that anywhere between zero to one to all of the polynucleotide analytes may be present in each droplet.

With regard to claim 10, the experiments in Han et al. are performed on DNA that is cell-free.

With regard to claims 11-12, Figure 5 of Vogelstein et al. teaches performing digital PCR on a stool sample of a subject to determine presence of cancer.

With regard to claim 20, the methods section of Vogelstein et al. teaches diluting the mixtures of polynucleotide analytes into a fixed concentration in the mixture of the plurality of droplets such that Figure 1 of Vogelstein et al. illustrates presence/absence patterns of the polynucleotide analyzes in each of the droplet wells.

It would have been obvious to someone of ordinary skill in the art at the time of the instant invention to modify the labeling of Han et al. by use of the labeling of Lin et al. wherein the motivation would have been that the barcodes of Lin et al. generate a combinatorial number of labels [Figure 2 of Lin et al.].

It would have been obvious to someone of ordinary skill in the art at the time of the instant invention to modify the labeling of Han et al. and Lin et al. by use of the digital PCR of Vogelstein wherein the motivation would have been that the partitioning into droplets facilitates identifying presence/absence patterns of polynucleotide analytes.

There would have been a reasonable expectation of success in combining Han et al., Lin et al., and Vogelstein et al. because all three studies analogously pertain to

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fluorophore labeling of polynucleotide analytes to determine their presence/absence patterns.

Response to arguments:

Applicant's arguments filed 23 May 2019 have been fully considered but they are not persuasive.

Applicant reiterates the limitations of the claims (specifically cumulative measurement) are argues, without support that Han et al. does not teach the limitations of the claims. This argument is not persuasive for the reasons discussed in the rejection statement above. In addition, the plots in Figure 5 of Han et al. represent cumulative measurements of quantum dots colors within the microbead.

Applicant reiterates the limitations of the claims (specifically the lack of immobilization) are argues, without support that neither Lin et al. nor Vogelstein et al. does not teach the limitations of the claims. This argument is not persuasive for the reasons discussed in the rejection statement above. In addition, Han et al. (and not Lin et al. or Vogelstein et al.) is relied upon for the teaching of the exclusion of immobilization and mass spectrometry.

The following rejection is reiterated:

35 U.S.C. 103 Rejection #2:

Claims 13 and 17-19 is/are rejected under pre-AIA 35 U.S.C. 103(a) as being unpatentable over Han et al. in view of Lin et al. in view of Vogelstein et al. as applied to

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claims 1-12 and 20 above, in further view of Empedocles et al. [US PGPUB

2002/0022273 A1; on IDS].

Claims 13 and 17-19 recite computer limitations pertaining the coding of polynucleotide analytes.

Han et al., Lin et al., and Vogelstein et al. make obvious using combinatorial hybridization and fluorophore labeling to identify presence/absence patterns of polynucleotide analytes, as discussed above.

Han et al., Lin et al., and Vogelstein et al. do not teach the computer limitations of the claims.

The document of Empedocles et al. studies differentiable spectral bar code methods and systems [title]. Figure 1 of Empedocles et al. illustrates the computer limitations of the claims.

It would have been obvious to someone of ordinary skill in the art at the time of the instant invention to modify the labeling of Han et al. and Lin et al. and the digital PCR of Vogelstein by use of the computers of Empedocles et al. wherein the motivation would have been that automation facilitates the execution of the method [Figure 1 of Empedocles et al.].

There would have been a reasonable expectation of success in combining Han et al., Lin et al., Vogelstein et al., and Empedocles et al. because all four studies analogously pertain to fluorophore labeling of polynucleotide analytes to determine their presence/absence patterns.

Response to arguments:

Applicant's arguments filed 23 May 2019 have been fully considered but they are not persuasive.

Applicant argues that Empedocles et al. does not overcome the alleged deficiencies of the initial obviousness prior art rejection. This argument is not persuasive because the combination of references in the instant prior art rejection makes obvious all of the limitations of the rejected claims.

The following rejection is reiterated:

35 U.S.C. 103 Rejection #3:

Claims 14-16 is/are rejected under pre-AIA 35 U.S.C. 103(a) as being unpatentable over Han et al. in view of Lin et al. in view of Vogelstein et al. in view of Empedocles et al. as applied to claims 1-13 and 17-20 above, in further view of Newton et al. [US PGPUB 2007/0072211 A1; on IDS].

Claims 14-16 recite having a health care professional make a clinical decision on cancer treatment based on the polynucleotide data.

Han et al., Lin et al., Vogelstein et al., and Empedocles et al. make obvious using combinatorial hybridization and fluorophore labeling to identify presence/absence patterns of polynucleotide analytes, as discussed above. Figure 5 of Vogelstein et al. conducts digital PCR on the stool of a subject to determine the presence of cancer.

Han et al., Lin et al., Vogelstein et al., and Empedocles et al. do not teach the clinical treatment of the claims.

The document of Newton et al. studies asymmetric PCR coupled with post-PCR characterization for the identification of nucleic acids [title]. Paragraphs 111 and 119 of

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Newton et al. teach that a health care professional makes clinical decisions on cancer treatment based on the nucleic acid data provided.

It would have been obvious to someone of ordinary skill in the art at the time of the instant invention to modify the labeling of Han et al. and Lin et al., the digital PCR of Vogelstein et al., and the computers of Empedocles et al. by use of the cancer treatment of Newton et al., wherein the motivation would have been that cancer treatment gives a physiological application to the combinatorial analysis of barcoded polynucleotide analytes [paragraphs 111 and 119 of Newton et al.].

There would have been a reasonable expectation of success in combining Han et al., Lin et al., Vogelstein et al., Empedocles et al., and Newton et al. because all five studies analogously pertain to analysis of polynucleotide analytes for biological applications.

Response to arguments:

Applicant's arguments filed 23 May 2019 have been fully considered but they are not persuasive.

Applicant argues that Empedocles et al. and Newton et al. do not overcome the alleged deficiencies of the initial obviousness prior art rejection. This argument is not persuasive because the combination of references in the instant prior art rejection makes obvious all of the limitations of the rejected claims.

E-mail Communications Authorization

Per updated USPTO Internet usage policies, Applicant and/or applicant's representative is encouraged to authorize the USPTO examiner to discuss any subject matter concerning the above application via Internet e-mail communications. See MPEP 502.03. To approve such communications, Applicant must provide written authorization for e-mail communication by submitting the following statement via EFS-Web (using PTO/SB/439) or Central Fax (571-273-8300):

Recognizing that Internet communications are not secure, I hereby authorize the USPTO to communicate with the undersigned and practitioners in accordance with 37 CFR 1.33 and 37 CFR 1.34 concerning any subject matter of this application by video conferencing, instant messaging, or electronic mail. I understand that a copy of these communications will be made of record in the application file.

Written authorizations submitted to the Examiner via e-mail are **NOT** proper. Written authorizations must be submitted **via EFS-Web** (using PTO/SB/439) **or Central Fax** (571-273-8300). A paper copy of e-mail correspondence will be placed in the patent application when appropriate. E-mails from the USPTO are for the sole use of the intended recipient, and may contain information subject to the confidentiality requirement set forth in 35 USC § 122. See also MPEP 502.03.

Conclusion

No claim is allowed.

THIS ACTION IS MADE FINAL. Applicant is reminded of the extension of time policy as set forth in 37 CFR 1.136(a).

A shortened statutory period for reply to this final action is set to expire **THREE MONTHS** from the mailing date of this action. In the event a first reply is filed within **TWO MONTHS** of the mailing date of this final action and the advisory action is not mailed until after the end of the **THREE-MONTH** shortened statutory period, then the

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shortened statutory period will expire on the date the advisory action is mailed, and any extension fee pursuant to 37 CFR 1.136(a) will be calculated from the mailing date of the advisory action. In no event, however, will the statutory period for reply expire later than SIX MONTHS from the mailing date of this final action.

Any inquiry concerning this communication or earlier communications from the Examiner should be directed to Russell Negin, whose telephone number is (571) 272-1083. This Examiner can normally be reached from Monday through Thursday from 10:30 am to 5 pm and variable hours on Fridays.

If attempts to reach the Examiner by telephone are unsuccessful, the Examiner's Supervisor, Marjorie Moran, Supervisory Patent Examiner, can be reached at (571) 272-0720.

Information regarding the status of the application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only.

/RUSSELL S NEGIN/
Primary Examiner, Art Unit 1631
17 July 2019